

Trino Deployment on Clusters

HPC Cluster Engineer Academy

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About Us



Lucian Chauvin
Texas A&M University



Jeremi Nuer
UC Santa Barbara

Your Data is Everywhere

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- Object Storage (S3)

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- SQL Databases

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- Even Google Sheets!

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- SQL Databases
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However, you want all your data to be easily accessible in one centralized location!

Trino is the Solution!

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- Query multiple sources (databases, data lakes, object storage)

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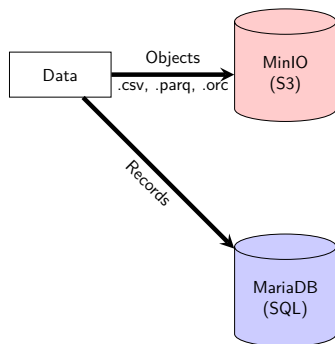
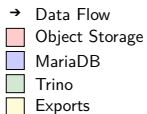
- Query multiple sources (databases, data lakes, object storage)
- Fast queries with an arbitrary amount of workers
- Highly scalable for big data environments with an unknown goal or end state

Architecture

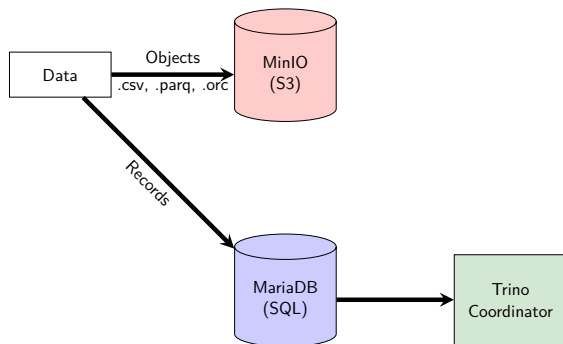
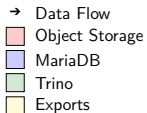
- Data Flow
-  Object Storage
-  MariaDB
-  Trino
-  Exports

Data

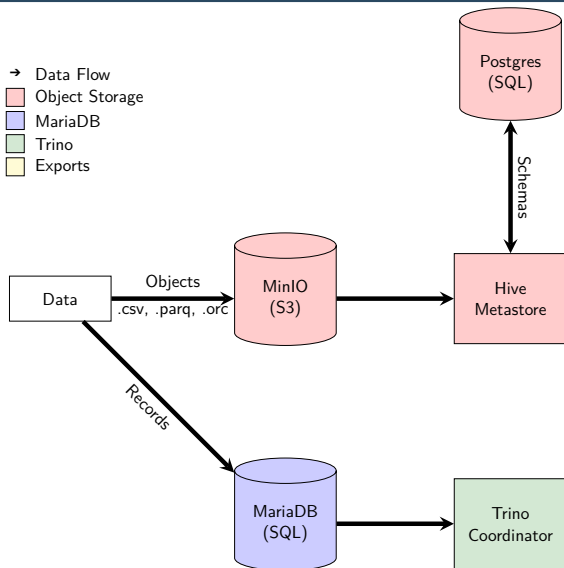
Architecture



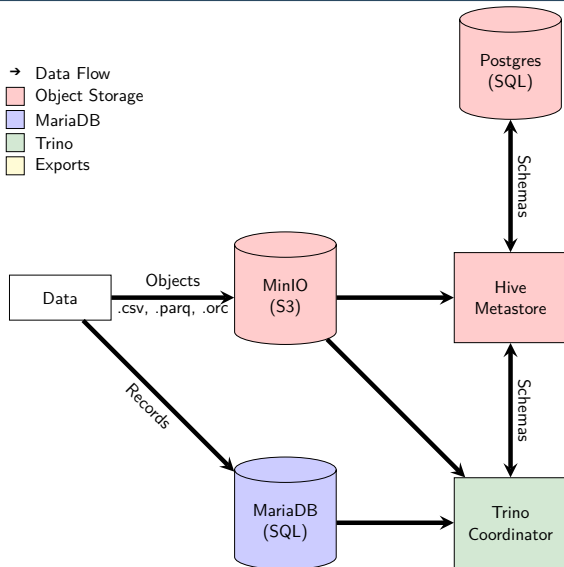
Architecture



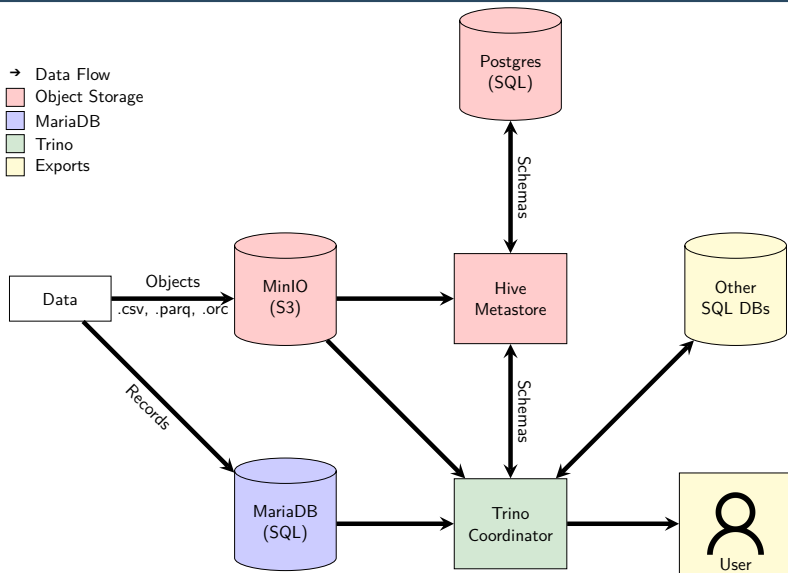
Architecture



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Architecture



Use Case

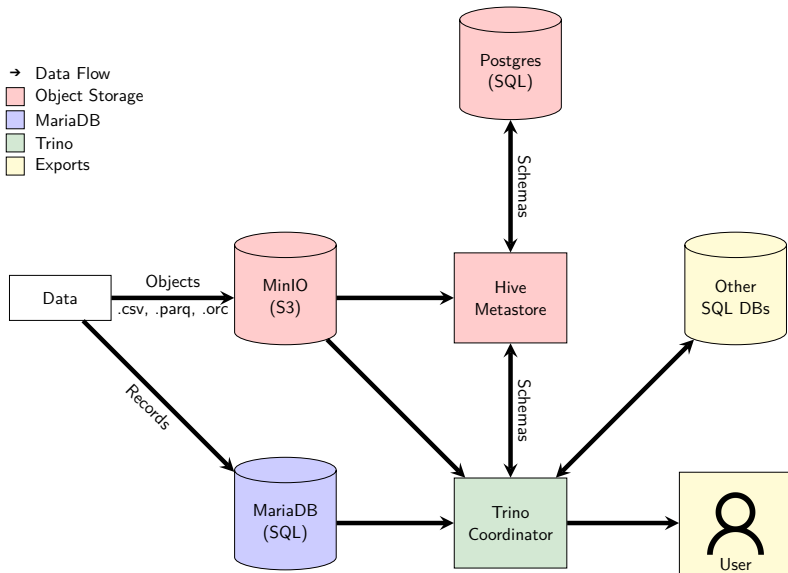
One use case that Trino enables is the ability to programmatically access, join, and write across datasources. Here is an example of this in practice:

id	first name	last name	age
7	Kellan	Sullivan	29
53	Kimberly	Grant	18
56	Kristian	Murray	22
70	Kellan	Rogers	22
85	Kelvin	Turner	19

(5 rows)

MinIO

Architecture



Use Case

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(5 rows)

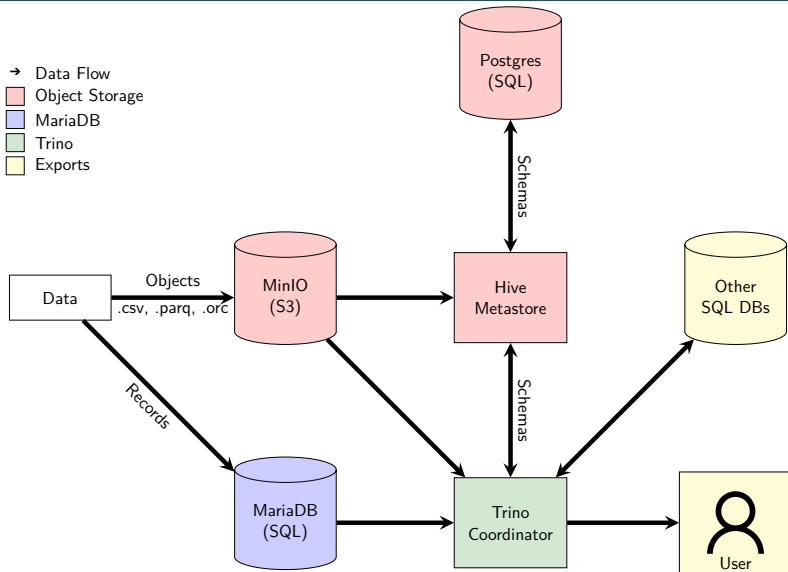
MinIO

id	label
7	10
53	10
56	10
70	10
85	10

(5 rows)

Other SQL DB

Architecture



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Other SQL DB

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(5 rows)

Joined

- Produced a fully automated Ansible script to install and connect Trino on a cluster
- Created a python script to programmatically interact with Trino

Both of these can be found in our CZ Gitlab remote here:

<https://lc.llnl.gov/gitlab/chauvin3/trino-on-cluster>

Challenges

- Hive Metastore installation
- Python bindings
- Database authorization across nodes using Ansible

Next Steps

- Deploy, diagnose, and compare Trino to its closed source competitors like Starburst
- Look into utilizing Trino in a ML workflow
- Test other Trino connectors

Sources

- MariaDB - <https://mariadb.com/kb/en/getting-installing-and-upgrading-mariadb/>
- MinIO - <https://min.io/docs/minio/linux/index.html>
- MinIO CLI (MC) - <https://min.io/docs/minio/linux/reference/minio-mc.html>
- Trino - <https://trino.io/docs/current/installation/deployment.html>
- trino-python-client - <https://github.com/trinodb/trino-python-client>

Made with $\text{\LaTeX}$

Thank you. Any questions?